THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Sample Test 1

1. (7 points) Define $\{a_n\}$ by $a_n = \dfrac{n-1}{n^2-3}$ for any positive integer n. Find the limit of $\{a_n\}$ using the ε definition.

2. Let X and Y be two nonempty bounded sets of real numbers. It is given that for any $x \in X$, there exists $y \in Y$ such that $x \leqslant y$.

   (a) (4 points) Prove that $\sup X \leqslant \sup Y$.

   (b) (3 points) Is it always true that $\inf X \leqslant \inf Y$?

3. Let $\{b_n\}$ be a sequence of real numbers. For each $k = 1, 2, 3, \ldots$, define $c_k = \inf\{b_1, b_2, \ldots, b_k\}$.

   (a) (1 point) Explain why each c_k is a real number.

   (b) (3 points) Given that $\{b_n\}$ is bounded, prove that $\{c_k\}$ is convergent.

   (c) (3 points) Is it always true that $\lim\limits_{k\to\infty} c_k = \liminf\limits_{n\to\infty} b_n$ if both are real numbers?

4. (7 points) Let $\{a_n\}$ and $\{b_n\}$ be two sequences of real numbers where a_1 and b_1 are positive. Suppose

$$a_{n+1} = a_n + b_n \quad \text{and} \quad b_{n+1} = a_n b_n$$

   for any positive integer n. Prove that the following are equivalent (that is, if any of them is true, then all of them are true).

   - $\{a_n\}$ is convergent
   - $\{b_n\}$ is convergent
   - $a_n \leqslant 1$ for any positive integer n

5. Prove or disprove each of the following. State clearly whether it is true or false.

   (a) (4 points) Let S be a bounded infinite set of real numbers. Then for any positive integer n, there exists $a \in S$ such that $\sup S - \dfrac{1}{n} < a < \sup S$.

   (b) (4 points) Let $\{a_n\}$ and $\{b_n\}$ be two sequences of real numbers. If $\{a_n\}$ has a subsequence with limit L_1 while $\{b_n\}$ has a subsequence with limit L_2, then $\{a_n + b_n\}$ has a subsequence with limit $L_1 + L_2$.

   (c) (4 points) Let $\{c_n\}$ be a sequence of real numbers such that the set $C = \{c_1, c_2, \ldots\}$ is a dense subset of $\mathbb{R}$ (that is, for any real numbers $x < y$, there exists $c \in C$ such that $x < c < y$). Then $\{c_n\}$ must be divergent.